Coordinatewise products on ℓ_1 are Lipschitz p-summing

Candidate full solution packet for arXiv:1805.02115

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Status. Candidate full solution, likely valid, pending human review. The theorem below answers the source question positively for every $1 \le p < \infty$ and proves an arity-independent strengthening.

1 Source question

Angulo-López and Fernández-Unzueta introduce Lipschitz p-summing multilinear operators in [1]. On page 6, immediately after Proposition 3.5, they ask whether the coordinatewise product

$$T : \ell_1 \times \ell_1 \longrightarrow \ell_1, \qquad T(x,y) = (x_j y_j)_{j\ge 1},$$

is Lipschitz p-summing. Here is the exact source passage.

So, in particular, this estimate holds for any finite set of vectors of the form $z_i = u_1^i \otimes \ldots \otimes u_n^i - v_1^i \otimes \ldots \otimes v_n^i \in \Sigma_{X_1,\ldots,X_n} - \Sigma_{X_1,\ldots,X_n}$. This gives rise to (i) in Theorem 3.1. To see the other relation, just observe that $T \in \mathcal{L}(X_1,\ldots,X_n;Y)$ is *strongly p-summing* exactly when T satisfies condition (i) in Theorem 3.1 for every $k \in \mathbb{N}$, $i = 1,\ldots,k$, $u_i \in X_1 \times \cdots \times X_n$ and $v_i = 0$.

Example 3.3 in [9] shows that $T \in \mathcal{L}(\ell_1,\ell_1;\ell_1)$, $T((x),(y)) = (x_j y_j)_j$ is a strongly p-summing multilinear operator whose associated linear mapping is not absolutely p-summing. We do not know if T is Lipschitz p-summing.

Corollary 3.6. *The Aron-Berner extension of a Lipschitz p-summing multilinear operator $T \in \Pi_p^{Lip}(X_1,\ldots,X_n;Y)$ takes its values in Y.*

Proof. By Proposition 3.5, T is strongly p-summing. Thus, by Theorem 2.2 in [14], its Aron-Berner extension is in $\mathcal{L}(X_1^{**},\ldots,X_n^{**};Y)$.
The statements stated below as propositions can be proved using standard arguments. We will only prove their corollaries. This type of results are usually known

Figure 1: Source page 6: the sentence posing the coordinatewise-product question, with the preceding difference-tensor context.

For completeness, an m-linear map $S : X_1 \times \cdots \times X_m \to Y$ is Lipschitz p-summing if a constant C exists such that, for every finite family $u_i, v_i \in X_1 \times \cdots \times X_m$,

$$\left(\sum_i \|S(u_i) - S(v_i)\|^p\right)^{1/p} \le C \sup_{\varphi \in B_{\mathcal{L}(X_1,\ldots,X_m)}} \left(\sum_i |\varphi(u_i) - \varphi(v_i)|^p\right)^{1/p}. \tag{1}$$

The least such C is denoted $\pi_p^{\mathrm{Lip}}(S)$.

2 Proof intuition

The definition only tests differences of two decomposable tensors. After flattening one tensor coordinate against all the others, such a difference is a matrix of rank at most two. Its coordinate diagonal has ℓ_1 norm at most the matrix trace norm, and rank two bounds that trace norm by $\sqrt{2}$ times the Hilbert–Schmidt norm.

The missing bridge to the right-hand side of (1) is supplied by random signs. A bounded sign array is a norm-one multilinear form on a product of ℓ_1 spaces. The lower Khintchine inequality recovers the Hilbert–Schmidt norm of each difference array from the L_p norm of its random-sign evaluation. Crucially, one first sums over the finite family and only then averages; the expectation is bounded by one common sign-array supremum, precisely the supremum allowed in (1).

3 The result

Let $\kappa_{p,\mathbb{K}} > 0$ be any lower scalar Khintchine constant over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, so that for every square-summable scalar family (a_α) and independent Rademacher signs (ε_α),

$$\left(\mathbb{E}\left|\sum_\alpha \varepsilon_\alpha a_\alpha\right|^p\right)^{1/p} \geq \kappa_{p,\mathbb{K}} \left(\sum_\alpha |a_\alpha|^2\right)^{1/2}. \tag{2}$$

Only positivity of this standard constant is used; see [3, 4], including the complex-scalar formulation.

Theorem 1. *Let $m \geq 2$ and $1 \leq p < \infty$. Over either scalar field, the coordinatewise product*

$$M_m : \underbrace{\ell_1 \times \cdots \times \ell_1}_{m \text{ times}} \longrightarrow \ell_1, \qquad M_m(x^1, \ldots, x^m) = \left(x_j^1 \cdots x_j^m\right)_{j \geq 1},$$

is Lipschitz p-summing, and

$$1 \leq \pi_p^{\mathrm{Lip}}(M_m) \leq \frac{\sqrt{2}}{\kappa_{p,\mathbb{K}}}.$$

In particular, the question on page 6 of [1] has an affirmative answer for every $1 \leq p < \infty$.

We isolate the rank-two observation.

Lemma 1. *Suppose*

$$z = x^1 \otimes \cdots \otimes x^m - y^1 \otimes \cdots \otimes y^m \in \ell_1(\mathbb{N}^m),$$

and set $\Delta z = (z_{j,\ldots,j})_{j \geq 1}$. Then

$$\|\Delta z\|_1 \leq \sqrt{2} \left(\sum_{\mathbf{j} \in \mathbb{N}^m} |z_{\mathbf{j}}|^2\right)^{1/2}. \tag{3}$$

Proof. Regard z as the matrix of an operator

$$Z : \ell_2(\mathbb{N}^{m-1}) \longrightarrow \ell_2(\mathbb{N})$$

by placing the first index in the row and the remaining indices in the column. Each decomposable tensor gives a rank-one operator under this flattening, so rank $Z \leq 2$. Moreover, Z is Hilbert–Schmidt and

$$\|Z\|_{S_2} = \left(\sum_{\mathbf{j}} |z_{\mathbf{j}}|^2\right)^{1/2}.$$

Choose scalars θ_j of modulus at most one so that $\theta_j z_{j,\dots,j} = |z_{j,\dots,j}|$, and let $Q : \ell_2(\mathbb{N}) \to \ell_2(\mathbb{N}^{m-1})$ be the diagonal partial isometry $Qe_j = \theta_j e_{(j,\dots,j)}$. Then $\|Q\| \le 1$, and trace duality gives

$$\|\Delta z\|_1 = |\operatorname{tr}(ZQ)| \le \|Z\|_{S_1}\|Q\| \le \|Z\|_{S_1}.$$

If s_1, s_2 are the only possibly nonzero singular values of Z, then

$$\|Z\|_{S_1} = s_1 + s_2 \le \sqrt{2}(s_1^2 + s_2^2)^{1/2} = \sqrt{2}\|Z\|_{S_2},$$

which proves (3). The argument applies directly to the infinite array because $z \in \ell_1(\mathbb{N}^m) \subset \ell_2(\mathbb{N}^m)$ and Z has finite rank. □

Proof of Theorem 1. For a sign array $\varepsilon = (\varepsilon_{\mathbf{j}})_{\mathbf{j}\in\mathbb{N}^m}$, define

$$\varphi_\varepsilon(x^1, \dots, x^m) := \sum_{\mathbf{j}=(j_1,\dots,j_m)\in\mathbb{N}^m} \varepsilon_{\mathbf{j}} x^1_{j_1} \cdots x^m_{j_m}.$$

The series is absolutely convergent, and

$$|\varphi_\varepsilon(x^1, \dots, x^m)| \le \prod_{r=1}^m \|x^r\|_1.$$

Because every coefficient has modulus one, $\|\varphi_\varepsilon\| = 1$.

Now fix finitely many pairs $u_i = (x_i^1, \dots, x_i^m)$ and $v_i = (y_i^1, \dots, y_i^m)$, and put

$$z_i = x_i^1 \otimes \cdots \otimes x_i^m - y_i^1 \otimes \cdots \otimes y_i^m.$$

The output difference is $M_m(u_i) - M_m(v_i) = \Delta z_i$. Lemma 1 and (2) yield

$$\|M_m(u_i) - M_m(v_i)\|_1^p \le \left(\frac{\sqrt{2}}{\kappa_{p,\mathbb{K}}}\right)^p \mathbb{E}\,|\langle \varepsilon, z_i\rangle|^p.$$

Summing over i before taking the supremum gives

$$\begin{aligned}\sum_i \|M_m(u_i) - M_m(v_i)\|_1^p &\le \left(\frac{\sqrt{2}}{\kappa_{p,\mathbb{K}}}\right)^p \mathbb{E}\sum_i |\varphi_\varepsilon(u_i) - \varphi_\varepsilon(v_i)|^p \\ &\le \left(\frac{\sqrt{2}}{\kappa_{p,\mathbb{K}}}\right)^p \sup_{\varphi\in B_{\mathcal{L}(\ell_1,\dots,\ell_1)}} \sum_i |\varphi(u_i) - \varphi(v_i)|^p.\end{aligned}$$

After taking pth roots, this is (1) with $C = \sqrt{2}/\kappa_{p,\mathbb{K}}$. Finally, $\pi_p^{\mathrm{Lip}}(M_m) \ge \|M_m\| = 1$, giving the stated two-sided bound. □

4 Verification and limitations

The included script checks 960 real/complex decomposable differences of arities two and three, verifies rank at most two and $\|\Delta z\|_1 \le \sqrt{2}\|z\|_2$, and performs exact finite sign enumeration for small families. The code is only a sanity check; the proof above is dimension-free and analytic.

This packet does not claim the exact value of $\pi_p^{\mathrm{Lip}}(M_m)$. It also does not claim the separate Section 7 question of [1]: Fernández-Unzueta explicitly answered that question in [2]. A bounded novelty search used the run indexes, exact arXiv phrase/title queries, and the local full text of later arXiv papers citing the source. No prior answer to the coordinatewise-product question or the rank-two/random-sign argument was found. Novelty confidence is moderate pending specialist review.

5 Human-review recommendation

The candidate should receive expert proof review. The highest-value checks are: (i) the trace pairing with the diagonal partial isometry in Lemma 1; (ii) the scalar Khintchine inequality over the complex field; and (iii) the order "sum, then average, then supremum" in the final display.

References

[1] J. C. Angulo-López and M. Fernández-Unzueta, *Lipschitz p-summing multilinear operators*, J. Funct. Anal. 279 (2020), no. 4, 108572; arXiv:1805.02115.

[2] M. Fernández-Unzueta, *Lipschitz p-summing multilinear operators correspond to Lipschitz p-summing operators*, arXiv:2204.01775 (2022).

[3] A. Khintchine, *Über dyadische Brüche*, Math. Z. 18 (1923), 109–116.

[4] J.-P. Kahane, *Some Random Series of Functions*, second ed., Cambridge Studies in Advanced Mathematics 5, Cambridge University Press, 1985.